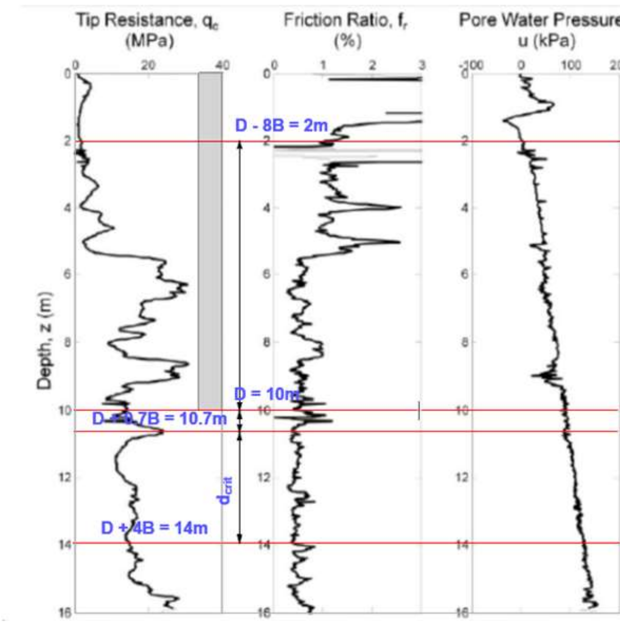
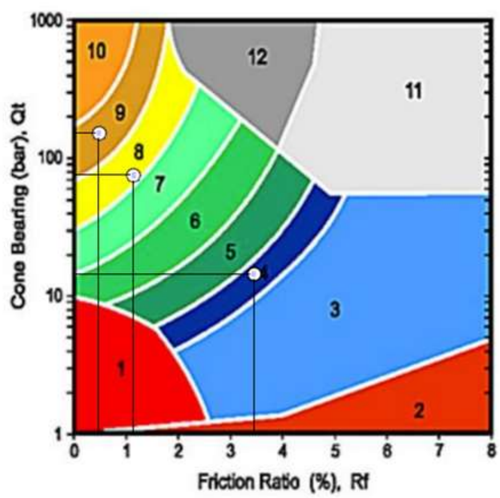
1) Determining the soil layers from CPT results



Layer	Depth	Soil Type	Avg qc	Avg fr
1	0-2 m	Sensitive clay / soft clay	1.5 MPa	3.5%
2	2-6 m	Silty sand / sand mixture	7.5 MPa	1.2%
3	6-10 m	Dense sand / silty sand	16.0 MPa	0.5%
4	10-13 m	Sandy silt / clayey silt	12.0 MPa	1.0%
5	13-16 m	Dense clean sand / overconsolidated silt	25.0 MPa	0.3%



ZONE		SBT
1		Sensitive, fine grained
2		Organic materials
3		Clay
4	1	Silty clay to clay
5		Clayey silt to silty clay
6		Sandy silt to clayey silt
7		Silty sand to sandy silt
8	2	Sand to silty sand
9	3	Sand
10		Gravely sand to sand
11		Very stiff fine grained*
12		Sand to clayey sand*

*over consolidated or cemented

2.a) Determining the bearing resistance.

$B := 1\ m$ $\quad D := 11\ m$ $\quad D_{eq} := B$ $\quad \lambda_b := 0.6$ $\quad \alpha_b := 0.7$ $\quad \alpha_{sq} := 0.55$ $\quad \mu_s := 1$ $\quad \mu_b := 0.9$

$q_{s.max.cohesive} := 80\ kPa$ $\quad q_{s.max.granular} := 120\ kPa$

$$A_b := 0.5 \cdot \frac{B^2 \cdot \pi}{4} \qquad d_{crit} := \frac{(D + 4 \cdot B) - (D + 0.7 \cdot B)}{m} = 3.3$$

$D + 4 \cdot B = 15\ m$ $\quad (D + 0.7 \cdot B) = 11.7\ m$ $\quad D - 8 \cdot B = 3\ m$

Average qc (KPa)

$q_{c.I} := 1500\ \text{kPa}$ $q_{c.II} := 7500\ \text{kPa}$ $q_{c.III} := 16000\ \text{kPa}$ here i choose lower qc so i get lower bearing capacity, just to be on the side of safety

Average fr (%)

$f_r := 3.5\ \%$ $f_r := 1.2\ \%$ $f_r := 0.5\ \%$

$$q_{c.I.mean} := \frac{1}{d_{crit}} \cdot \int_{D+4 \cdot B}^{D+0.7 \cdot B} q_{c.I}\, d z$$ $D + 4 \cdot B = 15\ \text{m}$ $(D + 0.7 \cdot B) = 11.7\ \text{m}$

$$q_{c.I.mean} := \frac{12000\ \text{kPa} \cdot 2.8 + 25000\ \text{kPa} \cdot 0.5}{2} = 23050\ \text{kPa}$$

$$q_{c.II.mean} := \frac{1}{d_{crit}} \cdot \int_{D+4 \cdot B}^{D+0.7 \cdot B} q_{c.II}\, d z$$ $D + 4 \cdot B = 15\ \text{m}$ $(D + 0.7 \cdot B) = 11.7\ \text{m}$

$$q_{c.II.mean} := \frac{12000\ \text{kPa} \cdot 2.8 + 25000\ \text{kPa} \cdot 0.5}{2} = 23050\ \text{kPa}$$

$$q_{c.III.mean} := \frac{1}{8 \cdot D_{eq}} \cdot \int_{D}^{D-8 \cdot B} q_{c.III}\, d z$$ $D := 9.5\ \text{m}$ $D - 8 \cdot B = 1.5\ \text{m}$

$$q_{c.III.mean} := \frac{1500\ \text{kPa} \cdot 0.5 + 7500\ \text{kPa} \cdot 4 + q_{c.III} \cdot 3.5}{3} = 28916.6667\ \text{kPa}$$

$$q_b := \lambda_b \cdot \alpha_b \cdot \frac{1}{2} \cdot \left(\frac{q_{c.I.mean} + q_{c.II.mean}}{2} + q_{c.III.mean} \right) = 10913\ \text{kPa}$$

$$R_{b.cal} := A_b \cdot q_b = 4285.5251\ \text{kN}$$

2.b) Determining the shaft resistance.

$P := B \cdot \pi = 3.1416\ \text{m}$

$$q_{s.1.cal} := 1.2 \cdot \mu_s \sqrt{\frac{q_{c.I}}{\text{kPa}}}\ \text{kPa} = 46.4758\ \text{kPa}$$ $\_ < \_ 80\ \text{kPa}$ pass

$$q_{s.2.cal} := 1.2 \cdot \mu_s \sqrt{\frac{q_{c.II}}{\text{kPa}}}\ \text{kPa} = 103.923\ \text{kPa}$$ $\_ > \_ 80\ \text{kPa}$ fail

becasue the qs cohesive maximam is 80, we can't go above it

$q_{s.2.cal} := 80\ \text{kPa}$

$q_{s.3.cal} := \alpha_{sq} \sqrt{\frac{q_{c.II}}{\text{kPa}}}\ \text{kPa} = 47.6314\ \text{kPa}$ $\quad$ _ <_ 80 kPa $\quad$ pass

$H_1 := 2\ \text{m}$ $\quad$ $H_2 := 4\ \text{m}$ $\quad$ $H_3 := 3.5\ \text{m}$

$A_{s.1} := H_1 \cdot \pi \cdot B = 6.2832\ \text{m}^2$ $\quad$ $A_{s.2} := H_2 \cdot \pi \cdot B$ $\quad$ $A_{s.3} := H_3 \cdot \pi \cdot B$

$$R_{s.cal} := \sum_{i=1}^{3} A_{s.i} \cdot q_{s.i}$$

$$R_{s.cal} := A_{s.1} \cdot q_{s.1.cal} + A_{s.2} \cdot q_{s.2.cal} + A_{s.3} \cdot q_{s.3.cal} = 1821.0603\ \text{kN}$$

$$R_{c.cal} := R_{s.cal} + R_{b.cal} = 6106.5854\ \text{kN}$$

$R_{c.cal.min} := R_{c.cal}$

$R_{c.cal.mean} := R_{c.cal}$

3) Ultimate Bearing Capacity

$n := 1$ $\quad$ $\zeta_{mean} := 1.40$ $\quad$ $\zeta_{min} := 1.40$

n	ξ_{mean}	ξ_{min}
1	1,40	1,40
2	1,35	1,27
3	1,33	1,23
4	1,31	1,20
5	1,29	1,15

$$R_{c.K} := \min\left(\left[\begin{matrix} \frac{\left(R_{c.cal.mean}\right)}{\zeta_{mean}} \\ \frac{\left(R_{c.cal.min}\right)}{\zeta_{min}} \end{matrix}\right]\right) = 4361.847\ \text{kN}$$

4) Design Bearing Capacity

$\gamma_t := 1.15$

$$R_{C.d} := \frac{R_{c.K}}{\gamma_t} = 3792.9102\ \text{kN}$$